

Problem set 5a: More practice of international trade

- 1) Suppose Italy has an hourly wage of 10 EUR per hour. Spain has an hourly wage of 15 EUR per hour. Furthermore, workers in Italy are twice as fast as workers in Spain. Each country can produce either Mangos or Trucks. In Italy, workers need 10 hours to produce one Mangos. In the same time, they can instead produce 2 Trucks. There is perfect competition and the wage is the same in both the Mango and Truck industry in both countries.
 - a. Will Italy export Trucks to Spain?
 - b. Italy decides to invest in a training program that will make its workers even faster, so that they will be 3 times (instead of just twice) as fast as in Spain. Will Italy export Trucks to Spain now?
- 2) Malta and Canada are trying to decide whether to impose a tariff. If both of them impose a tariff, they each have a payoff of 1. If only one country imposes a tariff, then that country gets 10 and the other country gets 0. If no country imposes a tariff, they each get 6.
 - a. Is this a prisoner's dilemma?
 - b. Does it matter whether they play this game 1 time or 50 times?
 - c. Does it matter whether they make the decision simultaneously, or whether one goes first and the other goes second (after observing the first)?
 - d. Suppose the countries are considering to cooperate by using a grim-trigger strategy. What is the critical discount factor above which cooperation (i.e. no one imposes a tariff) can be sustained as an equilibrium?
 - e. Suppose that the discount factor is $\delta = \frac{1}{3}$. Is cooperation then possible as an equilibrium?
 - f. Suppose that the discount factor is $\delta = \frac{9}{10}$. Will there be cooperation then for sure, or is there another equilibrium as well?
 - g. Suppose that the parties can bargain at zero transaction costs. Would the equilibrium then change, and if so – what will it be?